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Review



Ammonia inhibition in anaerobic digestion: A critical review of inhibition mechanisms and mitigation strategies

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ABSTRACT

Ammonia inhibition represents a significant bottleneck that constrains the stability and efficiency of anaerobic digestion (AD) processes, particularly for high-nitrogen organic waste. This review synthesizes multi-scale evidence to illustrate that while elevated ammonia levels can alter the overall microbial community structure, process failure is most directly linked to its profound inhibition of two thermodynamically constrained, low-redundancy metabolic steps: syntrophic propionate oxidation and acetoclastic methanogenesis. This selective vulnerability is fundamentally linked to the mechanism by which ammonia stress interferes with core cellular functions related to transcription and translation. To address this bottleneck, existing mitigation strategies can be categorized into those aimed at overcoming ammonia inhibition or those focused on ammonia removal. The former, which includes additive application and bioaugmentation, can alleviate ammonia stress in the short term and may further facilitate the acclimation of microbial communities, thereby supporting relatively stable operations under high-ammonia conditions. However, challenges related to cost, operational complexity, and long-term stability persist. Alternatively, ammonia removal strategies, such as iron-mediated anaerobic ammonium oxidation and vacuum evaporation, offer potential for fundamental mitigation, although techno-economic feasibility remains critical for practical implementation. Notably, the phenomenon of microaeration-driven nitrogen removal, repeatedly reported in recent studies, suggests that AD systems may possess an underappreciated micro-oxygen interface along with potentially unrecognized nitrogen transformation pathways. This observation indicates a potential paradigm shift from strictly anaerobic operation to controllable micro-oxygen regulation, representing a promising frontier for future research. This review aims to provide systematic insights into clarifying the cascading failure mechanisms of ammonia inhibition and developing next-generation, efficient, and economical mitigation strategies.

1. Introduction

With rapid socioeconomic development, global solid waste generation has continued to rise. The annual production of organic solid waste alone has reached 105 billion tonnes (Primmer, 2021). Anaerobic digestion (AD) presents an effective method for managing organic waste by enabling waste volume reduction and stabilization, thus contributing to the nationwide efforts to prevent and control pollution. Additionally, AD recovers renewable energy in the form of biogas, aiding in fossil fuel substitution and carbon emission mitigation (Hoy et al., 2023; Xing et al., 2022). Driven by the dual objectives of reducing pollution and carbon emissions, along with green development initiatives, AD has experienced rapid expansion in recent years (Jiang et al., 2022; Liu et al., 2021).

However, AD is susceptible to process instability. As illustrated in Fig. S1, ammonia inhibition is a primary trigger of AD disturbances and has garnered significant research attention among the various types of instability reported. This prominence largely arises from the low carbon-to-nitrogen (C/N) ratios (typically ranging from 3.2 to 16.3) found in common organic wastes, including food waste, kitchen waste, sewage sludge, and livestock manure (Di et al., 2023; Liu et al., 2021; Zhang et al., 2022). During the hydrolysis stage of AD, organic nitrogen is mineralized into ammonia. When ammonia nitrogen concentrations exceed 1400 to 5770 mg/L, microbial activity may be compromised,

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leading to acid accumulation, decreased biogas production, foaming, and other indicators of process deterioration (Kundu et al., 2023; Peng et al., 2021; Zhang et al., 2020). Reports indicate that ammonia inhibition can result in biogas yield losses of up to 30% to 60% in full-scale AD facilities (Nielsen and Angelidaki, 2008). Consequently, ammonia inhibition has attracted considerable interest. Early studies have extensively investigated inhibitory thresholds across diverse AD systems, and the mechanistic understanding has advanced with the development of molecular biological techniques. Concurrently, various mitigation strategies have been devised to reduce ammonia concentrations and enhance tolerance to ammonia stress, primarily including substrate pretreatment (Khanh Nguyen et al., 2021; Kim et al., 2024), co-digestion (Peng et al., 2022; Yamamoto-Ikemoto et al., 2023; Zhou et al., 2024), additive application (Di et al., 2023; Liu et al., 2024), bioaugmentation (Li et al., 2023, 2024; Yang et al., 2022), and ammonia stripping (Yellezuome et al., 2022). Recently, anaerobic ammonia removal technologies such as iron-mediated anaerobic ammonium oxidation (Feammox) and bioelectrochemical approaches have also demonstrated emerging potential (Wang et al., 2021; Yang et al., 2020, 2021a).

Building upon this extensive body of work, the present review aims to provide a systematic synthesis that addresses several key integrative challenges. First, while multi-omics studies have proliferated (Fuchs et al., 2018; Li et al., 2023; Yellezuome et al., 2022; Yenigün and Demirel, 2013), a coherent causal chain linking the primary biochemical perturbations induced by ammonia (e.g., at transcriptional and translational levels) to the selective inhibition of thermodynamically constrained metabolic steps, and ultimately to system-level failure, requires further articulation. Second, mitigation strategies are often discussed in a fragmented manner (Li et al., 2023; Romero-Güiza et al., 2016; Yellezuome et al., 2022). A unifying conceptual framework that distinguishes between overcoming inhibition biologically and removing ammonia physically or chemically is needed to clarify their applicability, technological maturity, and implementation barriers. Finally, promising emerging removal technologies—such as Feammox (Xia et al., 2023), vacuum evaporation (Khadir et al., 2024; Mustafa Abdelrahman et al., 2025), and particularly the novel micro-oxygen interface concept that bridges iron cycling with unexpected nitrogen loss (Liu et al., 2024)—require systematic analysis of their microbiological foundations and pathways toward controllable engineering processes.

To address these gaps, this review has three core objectives: (i) to integrate multi-scale evidence into a coherent mechanistic chain from cellular stress to process failure; (ii) to introduce and apply an "Overcome versus Removal" dichotomous framework for categorizing and evaluating mitigation strategies; and (iii) to focus on removal technologies with high engineering potential, elaborating on the scientific principles of the micro-oxygen interface by integrating iron-mediated ammonia oxidation with recent discoveries of nitrogen loss under micro-aerobic conditions, while identifying key research questions for future translation. This work seeks to provide a consolidated theoretical basis and innovative perspectives for developing next-generation strategies to mitigate ammonia inhibition.

2. Impacts of ammonia accumulation

As noted in the Introduction, substrates with low C/N, such as food waste and sewage sludge, which are rich in nitrogen, are the primary contributors to ammonia accumulation in AD systems. During the hydrolysis stage, organic nitrogen in these substrates is converted into inorganic nitrogen, primarily existing in the digestate as ammonium ions and free ammonia (FAN) (Peng et al., 2023). The equilibrium between these two forms is regulated by pH, temperature, and ionic strength, as illustrated in Equations (1) and (2). Due to its hydrophobic nature, FAN can diffuse freely across cell membranes into microbial cells and is widely recognized as the primary active form responsible for ammonia inhibition (Astals et al., 2018; Moerland et al., 2021).

$$FAN = \frac{f \cdot K_a \cdot TAN}{f \cdot K_a + 10^{-pH}} \quad (1)$$

$$\mathrm{Lg}\, f = -0.5 \cdot \left(\frac{\sqrt{I}}{1+\sqrt{I}} 0.1276 \cdot I \right) \quad (2)$$

where K_a represents the acid dissociation constant, with a value of 1.27×10^{-9} at 37 °C; f represents the activity coefficient; I denotes the ionic strength (mol/L), which was calculated by considering only NH_4^+ and its monovalent counterion (Zhang et al., 2022).

The detrimental effects of ammonia inhibition on AD systems can be characterized as a systemic functional disorder. This condition is primarily manifested through impaired microbial metabolic activity, which ultimately leads to a decline in performance and instability of the entire reactor. Literature data (see Table S1) indicate that the tolerance ranges of different AD systems to ammonia nitrogen vary considerably due to differences in substrate characteristics, operating conditions, and microbial communities. The inhibition thresholds for total ammonia nitrogen (TAN) and FAN can fluctuate within broad ranges of 1033-9000 mg/L and 34.4-200 mg/L, respectively. When ammonia concentrations fall within the inhibition threshold range, the system does not collapse immediately; rather, it may enter a dynamic equilibrium state known as the "inhibited steady state." (Chen et al., 2008). This state arises because methanogens are more sensitive to ammonia stress, leading to inhibited activity and a decreased consumption rate of volatile fatty acids (VFAs) (Hao et al., 2022). In contrast, the hydrolysis and acidogenesis processes are relatively less affected, resulting in continuous VFAs production (Ma et al., 2022). This imbalance between production and consumption triggers VFAs accumulation. However, the high concentration of ammonia nitrogen itself provides a strong endogenous alkalinity buffering capacity, which can neutralize the accumulated acidic products, thereby preventing catastrophic pH decline in the system (Chen et al., 2008; Zheng et al., 2021). Consequently, the system in the "inhibited steady state" exhibits significantly reduced but stable gas production efficiency, elevated VFAs concentrations, and relatively stable pH, representing a low-efficiency yet sustainable non-healthy operational state (Yu et al., 2025). Nevertheless, this balance has its limits. When the accumulation of ammonia nitrogen exceeds the system's buffering capacity (e.g., >5500 mg/L or even up to 9000 mg/L), the explosive increase of VFAs—literature reports indicate a surge of 158% to 325%, with concentrations exceeding 8000 mg/L—will overwhelm the buffering capacity of ammonia. This ultimately leads to a sharp decline in system pH (typically below 6.0), triggering irreversible inhibition of microbial communities, cessation of gas production, and complete acidification failure of the system. In such cases, tank cleaning and restart are often required (Table S1). The root cause of this macroscopic shift from "inhibited steady state" to "system collapse" lies in the profound toxic effects of ammonia on microbial cells and key metabolic pathways. The next section will systematically elaborate on the intrinsic inhibition mechanisms from the perspectives of enzyme activity, microbial communities, and multi-omics levels.

3. Multi-scale mechanisms of ammonia inhibition: from metabolic dysfunction to molecular origins

Traditional studies have often described ammonia inhibition at a singular scale (Fan et al., 2021; Liu et al., 2024; Peng et al., 2023; Zhang et al., 2022). To systematically elucidate its essence, this section develops a multiscale analytical framework that links macroscopic functional manifestations to microscopic molecular drivers. We first describe the differential enzymatic responses of key metabolic pathways under ammonia stress, analyze the microbial ecological strategies that drive these differences, and finally dissect the core molecular mechanisms responsible for specific functional deterioration (Gaspari et al., 2024; Liu et al., 2021, 2024; Wang et al., 2023). This integrated analysis indicates

that ammonia inhibition is not a uniform disturbance to the microbial community; rather, it represents a cascading failure that begins with the impairment of the cellular core machinery for transcription and translation, culminating in the selective breakdown of thermodynamically constrained pathways, including syntrophic propionate oxidation and acetoclastic methanogenesis (Feng et al., 2024; Liu et al., 2021; Peng et al., 2023; Zhang et al., 2022).

3.1. Enzymatic activity responses: identifying sensitive nodes in the metabolic network

AD is a complex bioprocess driven by a cascade of microbial enzymes (Yenigün and Demirel, 2013). Prior to the widespread adoption of meta-omics approaches, enzyme assay kits were predominantly utilized to quantify extracellular enzyme activities, serving as phenotypic indicators of metabolic function and process stability within AD systems (Hashemi et al., 2021; Thiel and Hattingh, 1967). Although this classical methodology lacks the precision of meta-omics in identifying specific taxa or genotypes responsible for particular activities, it offers a direct and effective means of assessing the overall metabolic state of the system under ammonia stress (Lebuhn et al., 2015). Evidence from multiple studies, as summarized in Table S1, reveals a consistent pattern that the impact of ammonia inhibition is highly selective across different stages of AD (Fan et al., 2021; Liu et al., 2024; Wang et al., 2018, 2022; Zheng et al., 2025a).

Overall, the enzyme activities during the hydrolysis and acidogenesis stages exhibit relative robustness, albeit with specific vulnerabilities. In the hydrolysis stage, α-glucosidase and phosphatase demonstrate strong ammonia tolerance, with activity decreases ranging from only 1%–7% even at TAN levels of 4000–5000 mg/L (Fan et al., 2021). However, protease presents a notable exception, as its activity can decline by 45.7%–88% at TAN levels around 2100 mg/L or when FAN is 150–210 mg/L (Liu et al., 2024; Zheng et al., 2025a). In the acidogenesis stage, butyrate kinase and propionate kinase typically exhibit only minor changes under moderate ammonia levels, with activity variations remaining below 10% (Liu et al., 2024; Wang et al., 2018). However, under extreme conditions, butyrate metabolism may be significantly disrupted, with butyrate kinase activity decreasing by approximately 84.5% at TAN levels exceeding 5000 mg/L (Wang et al., 2022).

In the acetogenesis stage, traditional enzyme-based assays are often inadequate due to the absence of a universal indicator enzyme that can accurately reflect the metabolic status of syntrophic communities, necessitating the use of omics-based analyses. Conversely, the methanogenesis stage is generally more sensitive and exhibits greater potential for acclimation. In non-acclimated systems, Liu et al. (2024) reported that a TAN concentration of approximately 2000 mg/L reduced the activity of coenzyme F420 by nearly 50%. Additionally, Wang et al. (2018) observed a 97% decrease in coenzyme F420 activity when FAN levels exceeded 200 mg/L. Furthermore, a TAN level of 5000 mg/L resulted in reductions in the activities of coenzyme M and cytochrome *c* by 96.6% and 78.8%, respectively (Wang et al., 2022), indicating a near-complete inhibition of electron transfer. However, full-scale engineering applications, such as Cambi™ plants, suggest that after prolonged acclimation, methanogens can sustain stable operations at TAN levels exceeding 2000 mg/L through community succession and metabolic regulation (Wett et al., 2014). This finding implies that under specific feeding regimes and extended environmental stress, methanogens may surpass the inhibition thresholds inferred from short-term enzymatic assays, highlighting their microbial adaptability (Zhang et al., 2022).

3.2. Microbial community succession: functional redundancy determines system resilience

Microbial communities respond to stress through population dynamics, where differences in functional redundancy serve as a key determinant of robustness across various metabolic processes (Zhang et al., 2020). Hydrolysis-acidogenesis bacteria typically exhibit high functional redundancy (Bertucci et al., 2019; Peng et al., 2021). When ammonia-sensitive genera, such as *Aminivibrio*, *Paludibacter*, and *Lentimicrobium* decline, functionally similar but more tolerant taxa, including *Aminobacterium*, *Petrimonas*, *Parabacteroides*, and *Prevotella,* can rapidly proliferate to occupy the vacated niches (Liu et al., 2021; Zhang et al., 2022). This process stabilizes the metabolic performance of the hydrolysis and acidogenesis stages.

In contrast, syntrophic communities responsible for VFA oxidation generally exhibit lower functional redundancy and higher susceptibility to ammonia inhibition compared to hydrolytic and acidogenic bacteria. This heightened vulnerability is linked to the limited diversity of key cultivated players and the time-sensitive nature of functional succession under stress. Regarding syntrophic acetate oxidation, only six syntrophic acetate-oxidizing bacteria (SAOB) have been successfully isolated in pure culture. These include the thermophilic, acetate-oxidizing, rod-shaped bacterium historically referred to as "*Reversibacter*" (Lee and Zinder, 1988; Plugge et al., 2010; Zinder and Koch, 1984), as well as *Thermacetogenium phaeum* (Hattori et al., 2000) and *Thermotoga lettingae* (Balk et al., 2002). Additionally, it encompasses mesophilic species such as *Schnuerera ultunensis* (Schnurer et al., 1996), *Syntrophaceticus schinkii* (Westerholm et al., 2010), and *Tepidanaerobacter acetatoxydans* (Westerholm et al., 2011b). Notably, ammonia tolerance varies significantly among these bacteria. Some mesophilic SAOBs (e.g., *Schnuerera ultunensis*, *Syntrophaceticus schinkii*) demonstrate considerable tolerance, sustaining growth at 8.4 – 14 g NH_4^+-N/L) (Schnurer et al., 1996; Westerholm et al., 2010, 2011b), while certain thermophilic SAOBs (e.g., *Thermacetogenium phaeum*) are highly sensitive, with growth rate inhibition observed at concentrations as low as 2.8 g NH_4^+-N/L (Kato et al., 2014). The historical isolate "*Reversibacter*" (lost before phylogenetic characterization) and *Thermotoga lettingae* have less clearly defined tolerance thresholds (Sun et al., 2014; Wang et al., 2015; Westerholm et al., 2011a). Similarly, the guild of cultivated syntrophic propionate-oxidizing bacteria (SPOB) is limited. This group includes four species of *Syntrophobacter* (Boone and Bryant, 1980; Chen et al., 2005; Harmsen et al., 1998; Wallrabenstein et al., 1995), two species of *Desulfotomaculum* (Nilsen et al., 1996; Plugge et al., 2002), three species of *Pelotomaculum* (de Bok et al., 2005; Imachi et al., 2007; Kosaka et al., 2006), *Smithella propionica* (Liu et al., 1999), and *Candidatus Desulfonatronobulbus propionicus* (Sorokin and Chernyh, 2016). Although recent metagenomic studies have identified candidate ammonia-tolerant SPOB lineages [e.g., *Candidatus Syntrophopropionicum ammoniitolerans* (Singh et al., 2021), *Desulfobulbaceae sp. MAG 45* (Singh et al., 2021), and *Candidatus Thermosyntrophopropionicum ammoniitolerans* (Singh et al., 2023)], their initial abundance in typical digesters is often low (Yi et al., 2020; Zhang et al., 2017). Therefore, during acute ammonia stress, the rapid inhibition of sensitive syntrophs may not be immediately compensated (Adams, 2024). The enrichment of tolerant but rare taxa and the re-establishment of syntrophic partnerships require a lag period, during which intermediates like propionate and acetate can accumulate critically, driving process destabilization (Zhang et al., 2020). This delayed functional resilience underscores why syntrophic oxidation steps frequently represent bottlenecks in system stability.

Methanogens exhibit functional differentiation and distinct tolerance profiles. *Methanosaeta* demonstrates a high affinity for acetate but is highly sensitive to ammonia, often experiencing significant suppression during the early stages of ammonia accumulation (Wang et al., 2025). In contrast, the metabolically versatile *Methanosarcina*, which is capable of both acetoclastic and hydrogenotrophic methanogenesis, along with hydrogenotrophic methanogens such as *Methanoculleus*, *Methanospirillum*, *Methanobacterium*, *Methanosphaerula*, and *Methanobrevibacter*, generally exhibit greater tolerance (Table S1) (Gaspari et al., 2024; Wang et al., 2023; Zhang et al., 2022). Under ammonia stress, methanogenic pathways typically shift from thermodynamically less favorable acetoclastic methanogenesis to hydrogenotrophic

methanogenesis. This shift is facilitated by *Methanosarcina* and SAOB converting acetate to H_2 and CO_2, which are subsequently utilized by hydrogenotrophic methanogens (Wang et al., 2015). Such functional compensation elucidates why the system can maintain gas production capacity under high ammonia conditions.

3.3. Molecular mechanisms: disrupted transcription and translation trigger cascading metabolic failure

Multi-omics studies, encompassing metagenomics, metatranscriptomics, and metaproteomics, have elucidated the molecular underpinnings of the aforementioned macroscopic phenomena. Initially, ammonia inhibition disrupts essential biological processes necessary for normal cellular function, subsequently compromising specific thermodynamically constrained metabolic pathways characterized by low functional redundancy.

3.3.1. Impairment of core cellular machinery

The expression of functional genes relies on the precise execution of transcription and translation, both of which are directly compromised by high-ammonia stress. Initially, transcription is impeded. Metatranscriptomic and metaproteomic evidence indicates that ammonia toxicity significantly downregulates the synthesis of key ribonucleic acid polymerase subunits (A′, D, M, α, and β′) in *Methanosarcina* (Feng et al., 2024; Liu et al., 2021). Notably, the expression of subunit M decreases from approximately 1000 fragments per kilobase of exon model per million mapped fragments to 0, signifying a severe impairment in the assembly of the transcription-initiation complex (Feng et al., 2024). Subsequently, translational capacity declines. Ammonia inhibition induces a broad downregulation of ribosomal large-subunit proteins (L3, L5, L6, L10, L11, L12, L13, L22, and L25) and small-subunit proteins (S3, S3Ae, S7, and S11) (Feng et al., 2024; Liu et al., 2021). The suppression is significantly stronger for the large subunit (Log_2FC −1.87 to −4.35) compared to the small subunit (−1.25 to −2.48). Additionally, the inhibited expression of aspartyl-transfer ribonucleic acid synthetase further constrains the supply of precursors, thereby disrupting the protein-synthesis chain (Liu et al., 2021). Moreover, energy limitations and increased stress on genome maintenance become evident. The expression of ATP synthase subunits (I, A, and D) decreases significantly by 10.2%–17.3%, weakening energy supply while concurrently inducing upregulation of deoxyribonucleic acid (DNA) repair proteins *LexA* and *RecA* to 278% and 158% of the control,

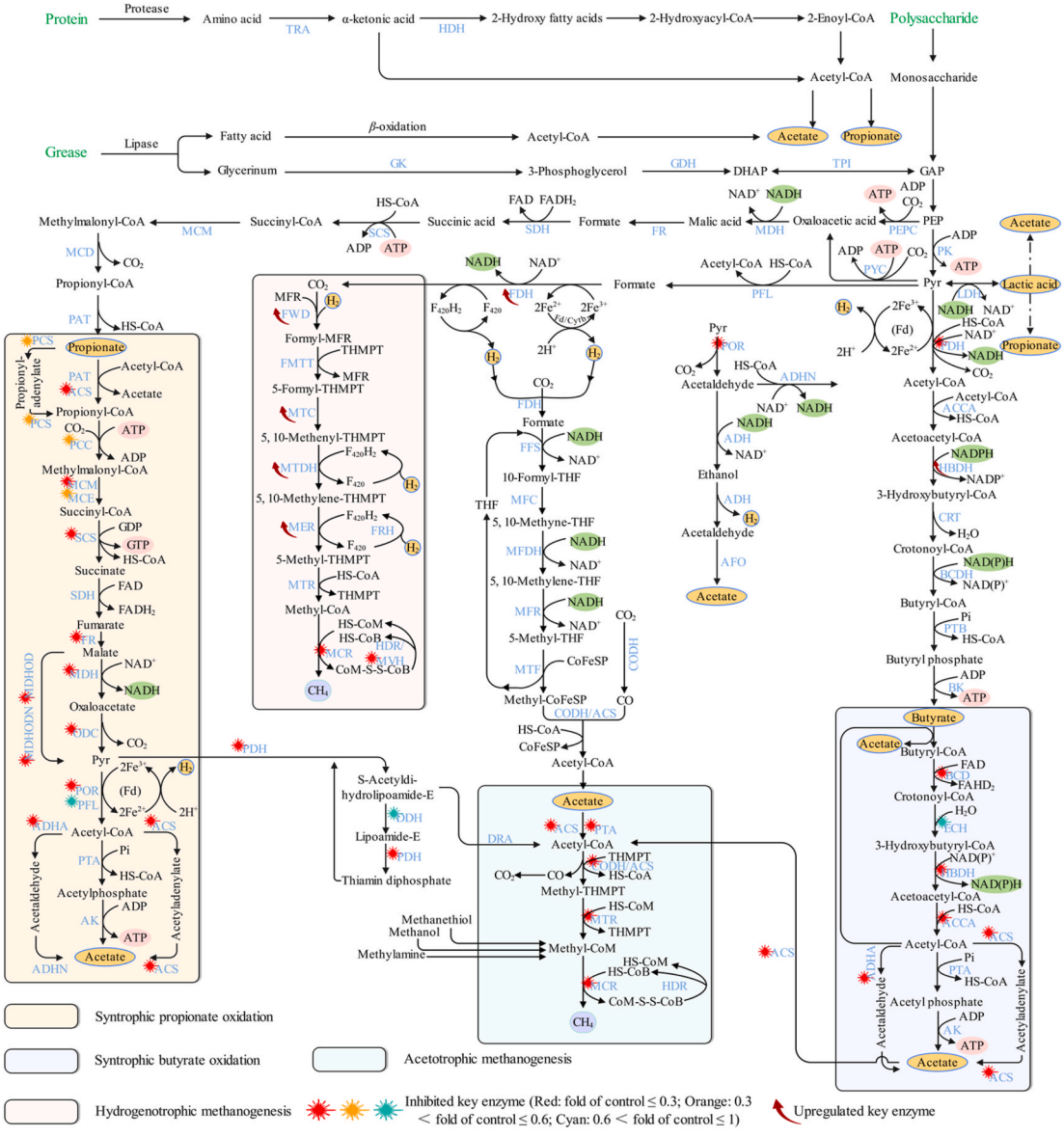


Fig. 1. Metabolic pathways across the four stages of AD, with abbreviations of functional enzymes provided in Table S2.

respectively (Feng et al., 2024). These patterns suggest severe DNA damage and a pronounced cellular stress response under high ammonia conditions.

3.3.2. Preferential collapse of key metabolic pathways

Damage to the core machinery described above is accompanied by significant declines in enzyme-system expression in metabolic pathways that require high energy and exhibit low functional redundancy. Fig. 1, a schematic compiled from literature data, visualizes the response spectrum of key metabolic enzymes under ammonia inhibition (Feng et al., 2024; Gaspari et al., 2024; Liu et al., 2021; Peng et al., 2023; Zhang et al., 2022). A quantitative legend is defined using the fold of control (FC), where red (FC $\leq$ 0.3), orange (0.3 < FC $\leq$ 0.6), and cyan indicate severe, moderate, and mild inhibition, respectively. In AD systems with TAN concentrations ranging from 1500 to 2100 mg/L, the transcription and expression levels of key butyrate-oxidation enzymes—BCD, HBDH, and ACCA—are only 7.4%–14.1% of the control (FC 0.074–0.141), indicating strong suppression of butyrate conversion to acetate (Feng et al., 2024; Liu et al., 2021). Furthermore, the inhibition of the propionate-metabolism pathway is even more severe, as illustrated in Fig. 1. The transcription and expression levels of key enzymes—PCS (FC 0.36), ACS (0.234–0.336), PCC (0.302), MCM (0.033–0.056), MCE (0.522), SCS (0.153–0.199), POR (0.149), MDHOD (0), and MDHODN (0)—are all markedly downregulated (Feng et al., 2024; Liu et al., 2021; Peng et al., 2023). Notably, under ammonia stress, metatranscriptomic analysis revealed no expression of MDHOD and MDHODN, which are key enzymes at the malate/oxaloacetate node of the methylmalonyl-CoA pathway. This finding indicates a near-complete transcriptional suppression of this functional module, resulting in a severe impairment of syntrophic propionate oxidation (Feng et al., 2024). Peng et al. (2023) further reported that at a TAN concentration of approximately 4000 mg/L, 85.7% (6/7) of the differentially expressed proteins related to propionate oxidation in Proteobacteria were significantly downregulated, including key enzymes such as FR and MDH. In contrast, butyrate oxidation exhibited pronounced functional resilience; although the dominant Firmicutes were inhibited, with approximately 63% of proteins downregulated, Proteobacteria maintained expression homeostasis, thereby sustaining degradation through niche compensation (Table S1). These observations indicate that inhibition of propionate metabolism is more fatal and less reversible than that of butyrate metabolism (Zhang et al., 2022).

During methanogenesis, gene expression profiles indicate a directional succession from acetoclastic methanogenesis to hydrogenotrophic and methylotrophic pathways. The acetoclastic pathway is inhibited first, as evidenced by the broad downregulation of transcription and protein expression levels of key enzymes, including ACS, PTA, CODH/ACS, MTR, and MCR (FC 0.062–0.687; Fig. 1). This downregulation indicates an impaired biological conversion of acetate to methane (Feng et al., 2024; Liu et al., 2021; Peng et al., 2023; Zhang et al., 2022). Under extreme conditions of 5000 mg NH_4^+-N/L, the expression of MCR in the obligate acetoclastic methanogen Methanothrix is significantly suppressed, suggesting that its metabolic activity has virtually ceased (Zhang et al., 2022). As the acetoclastic function declines, the system rapidly initiates a metabolic redirection towards syntrophic acetate oxidation and hydrogenotrophic methanogenesis. The upregulated expression of CODH/ACS complex subunits cdhCDE in mixotrophic methanogens suggests an enhancement of acetyl-CoA metabolism, which may be associated with a shift away from acetoclastic methanogenesis towards hydrogenotrophic methanogenesis and syntrophic acetate oxidation-coupled pathways under ammonia stress (Gaspari et al., 2024). In the hydrogenotrophic pathway, most key genes are upregulated, except for MCR and MVH (Fig. 1). Gaspari et al. (2024) further demonstrated that under ammonia stress (approximately 5000 mg NH_4^+-N/L), the catalytic role of MVH is completely supplanted at the transcriptional level by coenzyme F420-reducing hydrogenases. Given their capacity to generate $F420H_2$, this shift in electron transfer mode markedly reduces thermodynamic costs and constitutes a protective mechanism for sustaining biogas production under high ammonia conditions.

In summary, the primary effect of ammonia inhibition is a universal impairment of microbial cells' protein synthesis and energy metabolism capabilities. Building upon this foundation, inherent differences in functional redundancy among enzyme systems result in sustained hydrolysis and acidogenesis functions due to high redundancy. In contrast, two low-redundancy, high-energy-demanding key pathways—syntrophic propionate oxidation and acetoclastic methanogenesis—experience the most significant impact. These pathways undergo selective disruption at molecular, cellular, and population levels, ultimately leading to system-level phenotypes characterized by VFA accumulation and reduced gas production. Microbial communities mitigate this inhibition through functional redundant population replacement and metabolic pathway switching (e.g., towards hydrogenotrophic methanogenesis). However, the effectiveness of the former is extremely limited at the syntrophs level, rendering syntrophic propionate oxidation the most vulnerable link in the entire metabolic network.

4. Ammonia inhibition mitigation strategy

Following a systematic exposition of the multi-scale mechanisms underlying ammonia inhibition, the critical question arises: how can we effectively regulate this process to mitigate or eliminate its adverse effects? Addressing this question is essential for advancing the engineering implementation of AD. This section aims to establish a clear analytical framework for mitigation strategies, providing a critical synthesis and integration of existing technologies. As illustrated in Fig. 2, we categorize these strategies into two overarching paradigms based on the core logic of intervention: (1) strategies for overcoming ammonia inhibition, which aim to enhance the microbial community's adaptability to the high TAN environment or to provide alternative metabolic pathways and protection; and (2) strategies for ammonia removal, which aim to directly reduce ammonia concentration in the system through physical, chemical, or biological methods, thereby eliminating toxic stress at its source.

4.1. Strategies for overcoming ammonia inhibition

Supplementary Table S3 systematically summarizes the current strategies available for overcoming ammonia inhibition, which primarily include temperature control, additive application, co-digestion, bioaugmentation, and the application of multi-technology coupling. Although these strategies are theoretically feasible, their implementation in engineering practice often encounters challenges related to cost, stability, and operational complexity.

4.1.1. Temperature control

Temperature control serves as a strategy that indirectly influences the severity of ammonia inhibition by modifying physicochemical conditions. The fundamental rationale behind this approach is that temperature directly impacts the proportion of FAN (Eq. (1)), microbial metabolic rates, and community structure. While elevated temperatures can accelerate biochemical reactions, they also significantly increase FAN concentrations, exacerbating inhibition (Yenigün and Demirel, 2013). Recent studies have concentrated on the transition temperature range of 42–50 °C, striving to balance reaction rates with ammonia toxicity (He et al., 2024; Wang et al., 2024; Westerholm et al., 2015). Research indicates that operating at 45 °C can enhance methane yield by 7.8%–22.91% in ammonia-stressed AD systems (Chetawan et al., 2024; He et al., 2024; Wang et al., 2024). This enhancement in performance is frequently attributed to the enrichment of ammonia-tolerant *Methanosarcina* and the synergistic effects between acetoclastic and hydrogenotrophic methanogenesis pathways (He et al., 2024; Wang et al.,

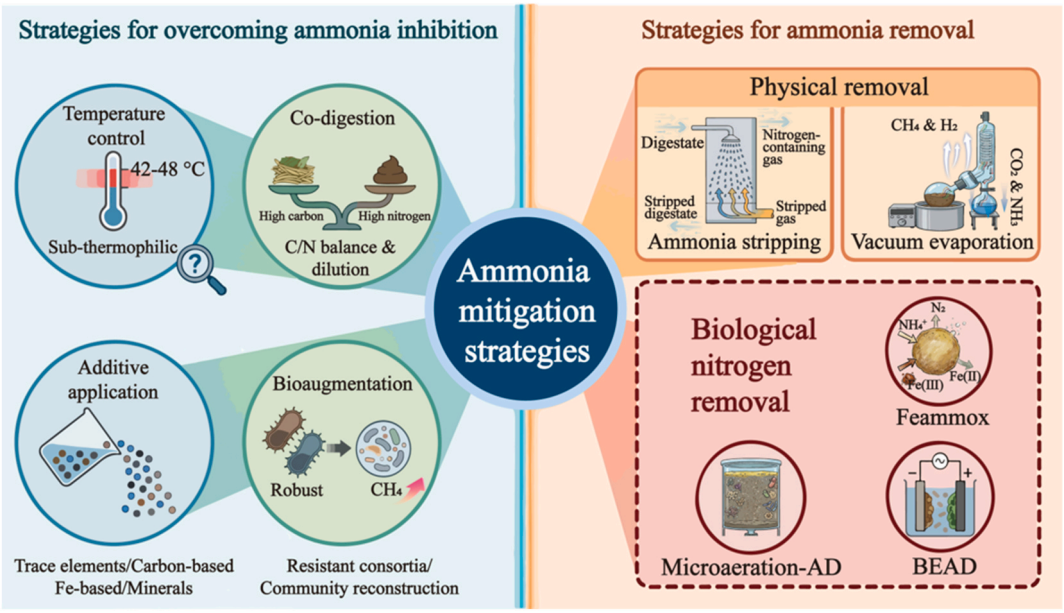


Fig. 2. Ammonia inhibition regulation technology.

2024). This finding aligns directly with the mechanistic discussion in Section 3 regarding methanogenic pathway plasticity and system stability, supported by functionally redundant microorganisms. However, the efficacy of the transition temperature strategy remains contentious. At slightly lower temperatures (e.g., 42 °C), studies have reported no significant increase in methane production, and even substantial accumulation of VFA, particularly propionate, in reactors with ammonium concentrations ranging from 5.4 to 5.8 g/L (Westerholm et al., 2015). This suggests that the success of this strategy is highly dependent on specific microbial acclimation and community assembly. Therefore, temperature control should not be regarded as a universal solution; its effectiveness is closely linked to its ability to foster a functional community with dedicated metabolic resilience.

4.1.2. Additive application

The additive-based strategy establishes a multi-dimensional defense against ammonia inhibition through physicochemical modulation, enhancement of electron transfer, and remodeling of the metabolic network. At the physicochemical level, materials such as Prussian blue analogue nanoparticles, allophane, and biochar leverage their high specific surface area and alkaline functional groups to mitigate the risk of acid-ammonia synergistic inhibition at its source by employing dual actions of TAN adsorption and VFA neutralization (Table S3). At the biological level, additives with electrochemical properties facilitate direct interspecies electron transfer (DIET) between methanogens and electroactive bacteria/syntrophs, including *Syntrophomonas*, *Sporanaerobacter*, *Thermovirga*, *Cloacibacillus*, *Defluviitoga*, and *Geobacter*, under high-ammonia conditions (3.0–12.24 g/L) (Cai et al., 2023; Wu et al., 2024; Yang et al., 2021; Zhang et al., 2021). This DIET-based interaction can circumvent hydrogen- and formate-mediated syntrophic pathways that may be inhibited at elevated ammonia levels, providing an efficient electron-transfer alternative and a thermodynamic solution to alleviate the vulnerability of syntrophic propionate oxidation (Table S3). Additionally, carbon- and iron-based materials can selectively enrich SAOB and ammonia-tolerant methanogens, steering the metabolic network from a single vulnerable mode toward a stable syntrophic acetate oxidation–hydrogenotrophic methanogenesis state (Table S3). Furthermore, nutrient elements (e.g., calcium ions) and organic promoters (e.g., glutathione) enhance the resilience of the micro-ecosystem by upregulating key enzyme activities (e.g., SCS) and reinforcing the structure of extracellular polymeric substances (Table S3). Following the application of these various classes of additives, methane production and the maximum methane yield have been reported to increase by 25%–350% and 106%–210%, respectively (Table S3). However, techno-economic feasibility and environmental sustainability remain significant obstacles to large-scale implementation. The effective dosages reported for current technologies typically range from 0.5% to 8%, with some instances exceeding 10%. For a facility treating 200 t/d, this translates into the annual consumption of several thousand tonnes of additives (Di et al., 2023; Liu et al., 2024; Wang et al., 2023). The high material costs, substantial logistical burdens, and potential long-term ecological accumulation risks associated with exogenous additives underscore the urgent need for future research to focus on developing modified materials that require lower dosages and exhibit high recyclability (Yellezuome et al., 2022; Yin et al., 2020).

4.1.3. Co-digestion

Co-digestion of high C/N waste with nitrogen-rich waste effectively dilutes the nitrogen concentration of the feedstock, thereby preventing excessive ammonia accumulation at the source (Peng et al., 2022; Zhang et al., 2022). A typical example involves the combination of straw and livestock manure (Guo et al., 2022; Jiang et al., 2022; Yamamoto-Ikemoto et al., 2023). An analysis of these cases reveals several challenges: straw and manure exhibit significant differences in their physicochemical properties; while manure is produced daily, straw is produced seasonally (Yamamoto-Ikemoto et al., 2023). These substantial differences in physicochemical properties necessitate the establishment of two distinct pretreatment lines for the substrates, leading to increased infrastructure investment costs (Kunatsa and Xia, 2022). To ensure continuous operation, seasonally produced straw requires not only adequate storage capacity but also appropriate preservation measures to prevent declines in its biogas potential during long-term storage. These requirements inevitably elevate operating costs and impose additional technical challenges on AD systems (Yellezuome et al., 2022). Currently, the feasibility of this technology is influenced more by regional feedstock synergies and supply chain management than by microbiological principles alone.

4.1.4. Bioaugmentation

Bioaugmentation directly addresses the functional deficiencies identified through mechanistic studies. For instance, the addition of syntrophic propionate-oxidizing bacteria, such as *Syntrophomonas* and

Syntrophobacter, aims to restore inhibited key metabolic bottlenecks, while the introduction of ammonia-tolerant methanogens seeks to enhance the functional redundancy of the system (Table S3). This strategy aligns closely with the conclusion in Section 3 that functional redundancy underpins system resilience. Wang et al. (2023) and Yang et al. (2022) reported that dosing an engineered bioaugmented consortium comprising *Syntrophomonas*, *Syntrophobacter*, *Tissierella, Methanosaeta*, *Syntrophaceticus*, and *Methanobrevibacter* effectively promoted propionate degradation and alleviated habitat toxicity caused by VFA accumulation, resulting in a 39%–49% increase in methane production. Furthermore, Wang et al. (2023) demonstrated that enriching hydrogenotrophic methanogens, including *Mathanosarcina*, *Methanoculleus*, and *Methanothermobacter*, significantly increased the gene abundances of the enzymes FMTT and MTC by 40.2% and 89.6%, respectively, potentially restoring methanogenic activity under ammonia inhibition from 0 to 184 mL g^{-1} VS (Table S3). However, challenges such as difficult colonization and susceptibility to washout remain primary barriers to the long-term effectiveness of this technology. Since community assembly in AD systems is driven by deterministic processes, exogenous microbial inoculants often struggle to compete with indigenous microorganisms and are unable to survive long-term in the harsh ammonia-stressed environment (Yang et al., 2022). Without improvements to the process matrix, such as pH and mass-transfer efficiency, bioaugmentation alone often provides only short-term relief and may even prove ineffective (Fuchs et al., 2018). To overcome this limitation, recent efforts have increasingly explored innovative approaches, including carrier-based immobilization to enhance biomass retention (Tang et al., 2022; Yang et al., 2022) and synthetic biology to integrate resistance genes into dominant indigenous strains (Rubin et al., 2021). These trends reflect an evolution from species-level reinforcement toward gene-level customization.

4.2. Strategies for ammonia removal

After systematically reviewing strategies aimed at enhancing system tolerance, we now turn our attention to the more fundamental paradigm of "ammonia removal". Unlike the previously discussed methods that rely on passive microbial adaptation, these technologies directly reduce ammonia nitrogen concentrations within the system through physical, chemical, or biological strategies, effectively alleviating inhibition at its source. As illustrated in Fig. 2, mainstream approaches include physicochemical separation techniques based on gas-liquid equilibrium, such as stripping and vacuum evaporation, alongside biological conversion methods that utilize emerging microbial metabolic pathways, including Feammox and bioelectrochemical systems.

4.2.1. Physicochemical separation technologies based on stripping and vacuum evaporation

Physicochemical technologies operate on the principle of ammonia gas-liquid equilibrium, creating conditions favorable for the volatilization of FAN to separate it from the liquid phase (Bi et al., 2020). Among these, ammonia stripping is a well-established process that involves introducing a gas stream into the digestate to carry away FAN (Baldi et al., 2018; Manzo et al., 2026). Its efficient operation typically relies on specific pH (usually >9) and temperature (>30 °C) conditions (Wu et al., 2024; Yu et al., 2025). However, the alkali addition required to achieve a high pH introduces additional costs, leads to the accumulation of alkali metals which can inhibit microbial activity, and may cause foaming (Yellezuome et al., 2022). The type of gas and the stripping gas flow rate are also critical operational parameters affecting process efficiency. Air stripping is cost-effective; however, residual oxygen may inhibit anaerobic microorganisms. In contrast, nitrogen stripping avoids oxygen toxicity and can also increase system pH by removing CO_2, although it is more expensive (Palakodeti et al., 2021). Furthermore, scaling in stripping towers and clogging caused by high-solids feeds present significant challenges that must be addressed in engineering practice. Consequently, stringent pretreatment of the influent and the development of unpacked tower configurations are often required. To date, only a limited number of systems have been implemented at pilot or full scale (Yellezuome et al., 2022).

Vacuum evaporation is an emerging technology for ammonia removal. It operates by utilizing negative pressure to lower the boiling point of the solution, allowing both water and FAN to evaporate simultaneously, followed by condensation to concentrate and recover ammonia (Khadir et al., 2024). A key advantage of this method is its ability to preferentially remove CO_2, which naturally raises the pH above 8.0, thereby facilitating efficient mass transfer and recovery of FAN without the need for external alkali addition (Haroun et al., 2022; Khadir et al., 2024). Despite the higher energy consumption associated with vacuum operation, this technology demonstrates significant life-cycle economic benefits, including cost reductions of approximately $5.3–10.5 million, due to savings in downstream nitrogen removal costs and sludge reduction (Mustafa Abdelrahman et al., 2025). In sidestream applications, it not only maintains ammonia nitrogen levels below inhibitory thresholds (<1.4 g/L) under high organic loading rates (6.1-11.3 kgCOD/m^3·d), but also supports stable methane production, with yields of approximately 0.214-0.25 L-CH_4/$gCOD_{fed}$ (Khadir et al., 2025a, 2026). Thus, vacuum evaporation effectively integrates ammonia inhibition mitigation with process intensification. However, the challenges associated with this technology include equipment corrosion resistance, scaling control, and ensuring stable long-term operation. Addressing these challenges is essential for enabling the widespread application of this innovative technology in the AD of high-nitrogen waste streams.

4.2.2. Biological conversion technologies based on feammox and its derivative pathways

Biological transformation technologies exploit the metabolic functions of specific microorganisms to convert ammonia nitrogen into harmless nitrogen gas, thereby offering significant potential for sustainability. Among these technologies, Feammox has garnered considerable attention in recent years. Feammox refers to the biological process in which microorganisms utilize Fe(III) as an electron acceptor to oxidize NH_4^+ to N_2 or NO_x^- under anaerobic conditions (Equations (3)–(5)) (Shi et al., 2024). The reaction pathway that yields N_2 as the end product is thermodynamically more favorable (Xia et al., 2022). However, in various systems, influenced by operational conditions, microbial communities, and habitat variations, the process can also generate nitrogenous products such as NO_2^- or NO_3^-, as documented in related case studies (Huang and Jaffé, 2015; Yang et al., 2021a). This process was first observed and confirmed in natural ecosystems such as wetlands (Shrestha et al., 2009). Its introduction into the field of AD stems from a dual expectation: on one hand, the Fe(III) reduction process itself can promote the degradation of complex organic matter and potentially enhance methanogenesis (Cruz Viggi et al., 2014); on the other hand, this pathway holds the promise of achieving synchronous nitrogen removal within the anaerobic digester (Yang et al., 2021b).

Early studies have regenerated Fe(III) by exogenously dosing iron oxides (e.g., magnetite, Fe_2O_3, and $Fe(OH)_3$) (Yang et al., 2018) or by combining nitrate/nitrite addition (Yang et al., 2020), thereby verifying the feasibility of Feammox in AD systems and achieving substantial nitrogen removal. For instance, intermittent dosing of NO_2^-/NO_3^- achieved nitrogen removal efficiencies of up to 98.6% and 90.1%, respectively; however, this also resulted in an approximate 76% decrease in methane production due to competition for electron donors (carbon sources) between denitrification and methanogenesis (Yang et al., 2020, 2021a). In contrast, Yuan et al. (2023) introduced Fe_2O_3 in a two-stage AD system, which not only activated a Feammox pathway driven by *Pseudomonas* and *Nitrososphaeraceae,* reducing ammonia concentrations by approximately 10.6%, but also reversed the energy-yield disadvantage, increasing bioenergy recovery by 208.43%. Recent studies have further focused on optimizing the forms of iron sources. Zheng et al. (2025c,

2025b) reported that, compared to highly crystalline hematite, amorphous ferrihydrite, due to its larger specific surface area, can be utilized more efficiently by the iron-reducing bacterium *Petrimonas*, maintaining FAN below 100 mg/L while increasing methane production by 56.3%–84.8%. Importantly, Zheng et al. (2025a) found that even in a pig-manure AD system characterized by high solids and high ammonia loading (sNH_3-N and sCOD of approximately 3.20 and 7.26 g/L, respectively), potassium ferrate pretreatment facilitated in situ generation of Fe(III) through the reduction of Fe(VI) from potassium ferrate, enriching iron-reducing bacteria such as *Clostridium* and *Dechloromonas*. This strategy successfully induced Feammox, alleviated ammonia inhibition by reducing NH_3-N from 3.0 g/L to 1.85–2.24 g/L, and increased methane production by 294.1%. These advancements indicate that Feammox-based technologies are gradually progressing beyond the initial trade-off of "sacrificing methane yield to achieve nitrogen removal" and evolving towards a new paradigm in which nitrogen removal and energy recovery can be enhanced simultaneously under high-ammonia conditions.

$$NH_4^+ + 3Fe(OH)_3 + 5H^+ \rightarrow 0.5N_2 + 3Fe^{2+} + 9H_2O \quad \triangle G = -245 \text{ kJ/mol} \quad (3)$$

$$NH_4^+ + 6Fe(OH)_3 + 10H^+ \rightarrow NO_2^- + 6Fe^{2+} + 16H_2O \quad \triangle G = -164 \text{ kJ/mol} \quad (4)$$

$$NH_4^+ + 8Fe(OH)_3 + 10H^+ \rightarrow NO_3^- + 8Fe^{2+} + 21H_2O \quad \triangle G = -207 \text{ kJ/mol} \quad (5)$$

Although novel iron-source strategies have improved process performance, the reliance on continuous exogenous dosing at the engineering application level continues to pose significant economic challenges (Xia et al., 2023).Estimates indicate that the removal of one tonne of ammonia nitrogen necessitates approximately 9.3 to 24.8 tonnes of iron reagents (Shi et al., 2024). Market surveys reveal that the associated treatment costs may reach as high as USD 225.00–6186.70. To address this economic bottleneck, research has increasingly focused on strategies that facilitate the recycling of iron reagents. The central concept involves re-oxidizing the Feammox product Fe(II) back to Fe (III). Given that nitrate dosing for iron regeneration has been shown to adversely affect biogas production, the introduction of trace oxygen to promote the chemical or biochemical oxidation of Fe(II) has emerged as the most promising in situ regeneration approach for engineering applications (Yang et al., 2020, 2021a). In systems with high concentrations of reductants, such as Fe(II) and organic matter, trace oxygen can be supplied to facilitate efficient iron cycling without disrupting the overall anaerobic environment. This strategy transforms Feammox from a "consumptive dosing process" into a "catalytic cycling process," representing a significant advancement toward engineering implementation (Yang et al., 2021b). Based on this iron redox cycle, intermittent aeration systems reported by Yang et al. (2021b) and Xia et al. (2023) achieved NH_4^+ removal efficiencies of 94.5% and 29.0%, respectively.

Notably, a deeper investigation into the strategy of "micro-oxygen regeneration" has revealed an even more striking phenomenon that extends beyond iron cycling itself. In AD systems without the addition of exogenous iron, controlled microaeration alone can lead to substantial and unexpected losses of ammonia nitrogen. This phenomenon has been consistently observed in multiple independent studies across different AD systems treating food waste and livestock manure (Liu et al., 2024; Yu et al., 2025). Specifically, in a food-waste AD system operated with continuous microaeration (4 mL air/min), Liu et al. (2024) reported a decrease in TAN by 75.60% compared to a strictly anaerobic control, while methane production increased by approximately 50%. In another study using cyclic microaeration (2 L/min) for substrate treatment, Yu et al. (2025) found that the system reduced ammonia nitrogen by 12.5% (with a total nitrogen loss of 14%), while leveraging a micro-oxygen environment to promote hydrolysis, ultimately increasing methane yield by 20%. These observations strongly suggest the existence of a distinctive, previously underappreciated micro-oxygen interface within AD systems. At this interface, trace amounts of oxygen may support an efficient microbial process for ammonia transformation that does not rely on exogenous Fe(III). This qualitative finding elevates "micro--oxygen" from a mere engineering tool for iron regeneration to a cutting-edge scientific concept with its own distinct microbial ecology and metabolic mechanisms.

However, translating "micro-oxygen" into a stable and controllable process presents fundamental challenges, primarily due to the stringent requirements for regulating dissolved oxygen concentrations. Excessive oxygen levels inhibit methanogenesis, and at an engineering scale, the heterogeneity in mixing and mass transfer makes it exceptionally difficult to maintain the "micro-oxygen" state that is easily achieved in laboratory reactors after scaling up. This difficulty partially explains why the positive outcomes reported at the laboratory scale are challenging to reproduce consistently in full-scale systems (Di et al., 2025; Khadir et al., 2025b; Ruan et al., 2019; Wei et al., 2024). Additionally, future research is urgently needed to identify the key functional microorganisms, core metabolic pathways, and ecological regulatory mechanisms operating under the "micro-oxygen interface," which form the theoretical foundation for transforming this incidental phenomenon into a reliable process.

4.2.3. Bioelectrochemical anaerobic digestion

Bioelectrochemical anaerobic digestion (BEAD) systems integrate electrochemical and biological processes. By embedding electrodes into an AD reactor, electrons released from the anodic oxidation of organic matter are transferred to the cathode through an external circuit, facilitating multiple reduction reactions, including those associated with ammonia oxidation and denitrification. This mechanism enables indirect ammonia removal (Liu et al., 2025). To date, ammonia removal in BEAD has primarily been attributed to three pathways. The first pathway is bioelectrochemical nitrification-denitrification, where electrical stimulation enriches the functional genes *amoA*, *hao*, and *nxrA*, thereby promoting the oxidation of ammonia and nitrite at the cathode (Chen et al., 2023). To address the oxygen requirement for ammonia oxidation, Chen et al. (2023) hypothesized that oxygen could be supplied via the influent or through weak oxygen evolution at the anode, although this has yet to be experimentally verified. Furthermore, a low-intensity electric field not only increases the abundance of denitrifiers (e.g., *SM1A02*, *Plasticicumulans*, *Ignavibacterium*, and *Methylopolylis*) and functional genes (*nasA*, *narGHI*, *narV*, *nirK*, *hcp*, and *nosZ*), thereby enhancing heterotrophic denitrification, but can also deliver electrons directly to cathodic electrotrophic denitrifiers such as *Thauera* (Chen et al., 2023; Huang et al., 2022; Liu et al., 2025). The second pathway is dissimilatory nitrate reduction to ammonium (DNRA), where electrode biofilms are enriched in key DNRA genes (*nrfA* and *nrfH*) that drive this reduction process (Chen et al., 2023; Wu et al., 2022). However, the precise sources of nitrite and nitrate remain unclear, necessitating isotope-based tracing in future research. The third pathway is electrochemically induced ammonia emission. Under applied voltage, solution ionization in the cathode chamber increases OH^- availability, which rapidly associates with NH_4^+ and promotes its release as NH_3 (Wang et al., 2021).

The relationship between ammonia removal and methanogenic performance in BEAD systems is nonlinear. Xu et al. (2024) reported that at an applied voltage of 0.6 V under an ammonia-inhibitory load of 5 g N/L, methane production and the maximum methane yield increased by 104.6% and 207.7%, respectively. These enhancements were attributed to improved syntrophic propionate metabolism and increased ammonia tolerance, rather than ammonia removal itself. Furthermore, Wang et al. (2021) demonstrated that as the applied voltage increased from 0 to 0.6 V, ammonia removal efficiency rose from 55% to 82%, while methane production declined by more than 48.9% due to the inhibition of methanogens. These findings suggest that the applied voltage is a critical factor in balancing nitrogen removal and methane production. However, BEAD studies are predominantly confined to laboratory settings, and the engineering implementation faces significant challenges. First, scaling up leads to increased ohmic losses, as larger electrode spacing at

the engineering scale markedly reduces energy efficiency (Janicek et al., 2014). Second, the capital costs are substantial, with proton exchange membranes priced at $1100/m^2, significantly increasing capital expenditure (Pant et al., 2011). Third, issues related to electrode scaling and fouling in complex matrices persist as a significant challenge.

5. Perspectives and challenges

Despite the increasingly deepened multi-scale understanding of ammonia inhibition and the growing diversity of mitigation strategies, the field continues to face a critical transition from merely "describing phenomena" to achieving "predictive control." To fundamentally enhance the AD of nitrogen-rich organic wastes under high-ammonia conditions, future research and practice should pursue paradigm innovation across three dimensions: mechanistic depth, technological integration, and engineering translation.

At the mechanistic level, current research has identified syntrophic propionate oxidation and acetoclastic methanogenesis as vulnerable links under ammonia inhibition. However, multi-omics datasets predominantly provide correlational evidence. Therefore, future work should focus on establishing a causal mechanistic chain that links molecular events to system failure. This includes employing synthetic biology tools to verify, in controlled systems, the direct consequences of impaired function of specific proteins, such as ribonucleic acid polymerase subunits, and quantifying how ammonia stress-induced reallocation of microbial energy influences the persistence priority of different metabolic pathways. Only through such efforts can the initial molecular targets that precipitate system collapse be identified, thereby providing a theoretical foundation for precise regulation grounded in microbial energetics.

At the technological level, no single stand-alone strategy is likely to simultaneously achieve high efficiency, cost-effectiveness, and long-term stability. Future breakthroughs are expected to arise from the development of synergistic composite solutions that incorporate multiple principles. For instance, ammonia-adsorbing materials could be combined with conductive media that facilitate iron cycling or DIET, thereby enabling an integrated additive system characterized by "adsorption-bioconversion-electron enhancement." Particularly noteworthy is the proactive construction and regulation of the micro-oxygen interface, which represents a paradigm shift from strictly anaerobic operations to controllable micro-oxygen regulation. The key to transforming this concept into a viable technology lies in elucidating the symbiotic network of functional microorganisms, such as potential aerobic ammonia-oxidizing bacteria, heterotrophic denitrifiers, and aerobic-tolerant methanogens, within this interface. Additionally, defining the precise thresholds of dissolved oxygen, pH, and redox potential necessary to maintain their dynamic balance is crucial. This understanding will provide the foundational design basis for achieving gradient and precise oxygen delivery in engineering implementations.

At the engineering-translation level, it is essential to acknowledge and systematically address the overarching challenges that arise when laboratory advances are scaled to full-scale implementation. This necessitates rigorous life-cycle sustainability assessments of each strategy, particularly those involving exogenous additives such as iron reagents and functional materials, along with prioritizing circular-economy pathways in which functional materials are produced from waste-derived resources. Meanwhile, given the complexity and variability of practical engineering feedstocks, it is crucial to develop control systems that integrate online monitoring with intelligent feedback. Future AD systems should possess the capability to sense real-time changes in key parameters, such as FAN, VFAs, and hydrogen, allowing for dynamic adjustments in operational strategies, including feedstock blending ratios, sidestream treatment intensity, and micro-aeration periods. Such capabilities would facilitate a transition from a "post-failure repair" approach to a "pre-inhibition intervention" strategy.

In summary, overcoming the bottleneck of ammonia inhibition requires deep, cross-scale synergy that spans mechanistic understanding at the microscopic level, the creation of composite technologies, and intelligent engineering control. The ultimate vision is to develop next-generation AD processes that can self-sense, dynamically adapt, and be precisely optimized. This not only represents an engineering imperative for enhancing the efficiency of organic waste treatment and energy recovery but also provides an excellent research paradigm for addressing the frontier fundamental challenge of achieving precise functional regulation in complex microbial ecosystems.

6. Conclusion

This review systematically demonstrates that the core of ammonia inhibition–induced destabilization in anaerobic digestion lies in the selective inhibition of two thermodynamically constrained and low-redundancy metabolic nodes: syntrophic propionate oxidation and acetoclastic methanogenesis. To address this bottleneck, existing regulatory strategies can be categorized into two fundamentally different paradigms: those aimed at overcoming ammonia inhibition and those focused on ammonia removal. This review indicates that overcoming strategies, such as additive application and bioaugmentation, can enhance microbial acclimation, thereby supporting relatively stable operations under high-ammonia conditions. However, the cost and long-term effectiveness of these strategies remain critical considerations. In contrast, removal-oriented strategies, such as vacuum evaporation and Feammox, offer more fundamental potential but face challenges related to techno-economic feasibility. Notably, the frequently observed phenomenon of micro-oxygen-driven nitrogen removal in recent studies suggests that AD systems may possess an underappreciated micro-oxygen interface with associated nitrogen metabolic potential. This insight marks a potential paradigm shift from strict anaerobic operation toward controllable micro-oxygen regulation. However, before this concept can be translated into a reliable technology, the key microorganisms, metabolic pathways, and precise control approaches operating at this interface must be elucidated. In the future, cross-scale solutions that integrate multiscale mechanistic understanding, smart materials, and real-time control are likely to be essential for achieving efficient and cost-effective mitigation of ammonia inhibition.

CRediT authorship contribution statement

Linpei Han: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Lei Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Jiaqi Huang:** Visualization, Methodology. **Wenjie Ye:** Visualization, Conceptualization. **Yongli Liu:** Visualization, Methodology. **Bin Zhan:** Visualization, Methodology. **Yilu Gu:** Visualization, Methodology. **Xuya Peng:** Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

The authors do not have permission to share data.

References

Adams, M., 2024. Ammonia-stressed anaerobic digestion: sensitivity dynamics of key syntrophic interactions and methanogenic pathways-A review. J. Environ. Manag. 371, 123183. https://doi.org/10.1016/j.jenvman.2024.123183.

Astals, S., Peces, M., Batstone, D.J., Jensen, P.D., Tait, S., 2018. Characterising and modelling free ammonia and ammonium inhibition in anaerobic systems. Water Res. 143, 127–135. https://doi.org/10.1016/j.watres.2018.06.021.

Baldi, M., Collivignarelli, M.C., Abbà, A., Benigna, I., 2018. The valorization of ammonia in manure digestate by means of alternative stripping reactors. Sustainability 10, 3073. https://doi.org/10.3390/su10093073.

Balk, M., Weijma, J., Stams, A.J.M., 2002. Thermotoga lettingae sp. nov., a novel thermophilic, methanol-degrading bacterium isolated from a thermophilic anaerobic reactor. Int. J. Syst. Evol. Microbiol. 52, 1361–1368. https://doi.org/10.1099/00207713-52-4-1361.

Bertucci, M., Calusinska, M., Goux, X., Rouland-Lefèvre, C., Untereiner, B., Ferrer, P., Gerin, P.A., Delfosse, P., 2019. Carbohydrate hydrolytic potential and redundancy of an anaerobic digestion microbiome exposed to acidosis, as uncovered by metagenomics. Appl. Environ. Microbiol. 85. https://doi.org/10.1128/AEM.00895-19.

Bi, S., Qiao, W., Xiong, L., Mahdy, A., Wandera, S.M., Yin, D., Dong, R., 2020. Improved high solid anaerobic digestion of chicken manure by moderate in situ ammonia stripping and its relation to metabolic pathway. Renew. Energy 146, 2380–2389. https://doi.org/10.1016/j.renene.2019.08.093.

Boone, D.R., Bryant, M.P., 1980. Propionate-degrading bacterium, *Syntrophobacter wolinii* sp. nov. gen. nov., from methanogenic ecosystems. Appl. Environ. Microbiol. 40, 626–632. https://doi.org/10.1128/aem.40.3.626-632.1980.

Cai, Y., Shen, X., Meng, X., Zheng, Z., Usman, M., Hu, K., Zhao, X., 2023. Syntrophic consortium with the aid of coconut shell-derived biochar enhances methane recovery from ammonia-inhibited anaerobic digestion. Sci. Total Environ. 872, 162182. https://doi.org/10.1016/j.scitotenv.2023.162182.

Chen, S., Liu, X., Dong, X., 2005. *Syntrophobacter sulfatireducens* sp. nov., a novel syntrophic, propionate-oxidizing bacterium isolated from UASB reactors. Int. J. Syst. Evol. Microbiol. 55, 1319–1324. https://doi.org/10.1099/ijs.0.63565-0.

Chen, Y., Cheng, J.J., Creamer, K.S., 2008. Inhibition of anaerobic digestion process: a review. Bioresour. Technol. 99, 4044–4064. https://doi.org/10.1016/j.biortech.2007.01.057.

Chen, Y., Dai, X., Wang, H., Liu, X., Gu, L., 2023. Insights into the mitigative effects of microbial electrolysis cell on anaerobic digestion under high ammonia condition. Chem. Eng. J. 474, 145760. https://doi.org/10.1016/j.cej.2023.145760.

Chetawan, W., Krishnan, S., Saritpongteeraka, K., Palamanit, A., Gabriel, D., Chaiprapat, S., 2024. Elucidating the role of sub-thermophilic temperature and pre-hydrolyzation for effective upgrading scheme of old swine manure digesters. Bioresour. Technol. 408, 131199. https://doi.org/10.1016/j.biortech.2024.131199.

Cruz Viggi, C., Rossetti, S., Fazi, S., Paiano, P., Majone, M., Aulenta, F., 2014. Magnetite particles triggering a faster and more robust syntrophic pathway of methanogenic propionate degradation. Environ. Sci. Technol. 48, 7536–7543. https://doi.org/10.1021/es5016789.

de Bok, F.A.M., Harmsen, H.J.M., Plugge, C.M., de Vries, M.C., Akkermans, A.D.L., de Vos, W.M., Stams, A.J.M., 2005. The first true obligately syntrophic propionate-oxidizing bacterium, *Pelotomaculum schinkii* sp. nov., co-cultured with *Methanospirillum hungatei*, and emended description of the genus pelotomaculum. Int. J. Syst. Evol. Microbiol. 55, 1697–1703. https://doi.org/10.1099/ijs.0.02880-0.

Di, L., Wang, F., Li, S., Wang, H., Zhang, D., Yi, W., Shen, X., 2023. Influence of nano-Fe_3O_4 biochar on the methanation pathway during anaerobic digestion of chicken manure. Bioresour. Technol. 377, 128979. https://doi.org/10.1016/j.biortech.2023.128979.

Di, Y., Hou, W., Bian, C., Zheng, T., Xiao, B., Li, L., 2025. Enhancing anaerobic digestion of swine manure using microbial electrolysis cell and microaeration. Chem. Eng. J. 514, 163319. https://doi.org/10.1016/j.cej.2025.163319.

Fan, Y., Yang, X., Lei, Z., Zhang, Z., Kobayashi, M., Adachi, Y., Shimizu, K., 2021. Alleviation of ammonia inhibition via nano-bubble water supplementation during anaerobic digestion of ammonia-rich swine manure: buffering capacity promotion and methane production enhancement. Bioresour. Technol. 333, 125131. https://doi.org/10.1016/j.biortech.2021.125131.

Feng, K., Lian, S.-J., Zou, H., Guo, R.-B., Dai, X.-F., Fu, S.-F., 2024. Ammonia stress on anaerobic digestion: the more sensitive propionic/butyric acid-degrading bacteria over methanogens. ACS ES&T Eng. 4, 627–638. https://doi.org/10.1021/acsestengg.3c00411.

Fuchs, W., Wang, X., Gabauer, W., Ortner, M., Li, Z., 2018. Tackling ammonia inhibition for efficient biogas production from chicken manure: status and technical trends in Europe and China. Renew. Sustain. Energy Rev. 97, 186–199. https://doi.org/10.1016/j.rser.2018.08.038.

Gaspari, M., Ghiotto, G., Centurion, V.B., Kotsopoulos, T., Santinello, D., Campanaro, S., Treu, L., Kougias, P.G., 2024. Decoding microbial responses to ammonia shock loads in biogas reactors through metagenomics and metatranscriptomics. Environ. Sci. Technol. 58, 591–602. https://doi.org/10.1021/acs.est.3c07840.

Guo, H.-G., Li, Q., Wang, L.-L., Chen, Q.-L., Hu, H.-W., Cheng, D.-J., He, J.-Z., 2022. Semi-solid state promotes the methane production during anaerobic co-digestion of chicken manure with corn straw comparison to wet and high-solid state. J. Environ. Manag. 316, 115264. https://doi.org/10.1016/j.jenvman.2022.115264.

Hao, T., Xiao, Y., Varjani, S., 2022. Transiting from the inhibited steady-state to the steady-state through the ammonium bicarbonate mediation in the anaerobic digestion of low-C/N-ratio food wastes. Bioresour. Technol. 351, 127046. https://doi.org/10.1016/j.biortech.2022.127046.

Harmsen, H.J.M., Van Kuijk, B.L.M., Plugge, C.M., Akkermans, A.D.L., de Vos, W.M., Stams, A.J.M., 1998. *Syntrophobacter fumaroxidans* sp. nov., a syntrophic propionate-degrading sulfate-reducing bacterium. Int. J. Syst. Bacteriol. 48, 1383–1387. https://doi.org/10.1099/00207713-48-4-1383.

Haroun, B., Bahreini, G., Zaman, M., Jang, E., Okoye, F., Elbeshbishy, E., Santoro, D., Walton, J., Al-Omari, A., Muller, C., Bell, K., Nakhla, G., 2022. Vacuum-enhanced anaerobic fermentation: achieving process intensification, thickening and improved hydrolysis and VFA yields in a single treatment step. Water Res. 220, 118719. https://doi.org/10.1016/j.watres.2022.118719.

Hashemi, S., Hashemi, S.E., Lien, K.M., Lamb, J.J., 2021. Molecular microbial community analysis as an analysis tool for optimal biogas production. Microorganisms 9, 1162. https://doi.org/10.3390/microorganisms9061162.

Hattori, S., Kamagata, Y., Hanada, S., Shoun, H., 2000. *Thermacetogenium phaeum* gen. nov., sp. nov., a strictly anaerobic, thermophilic, syntrophic acetate-oxidizing bacterium. Int. J. Syst. Evol. Microbiol. 50, 1601–1609. https://doi.org/10.1099/00207713-50-4-1601.

He, Huiban, Wang, Z., Wang, W., He, Haoxing, Yan, J., Wang, H., Cui, Z., Yuan, X., 2024. Mitigating short-circuits through synergistic temperature and hydraulic retention time control for enhancing methane yield in continuous stirred-tank reactors. Energy 289, 129914. https://doi.org/10.1016/j.energy.2023.129914.

Hoy, Z.X., Woon, K.S., Chin, W.C., Van Fan, Y., Yoo, S.J., 2023. Curbing global solid waste emissions toward net-zero warming futures. Science 382, 797–800. https://doi.org/10.1126/science.adg3177.

Huang, S., Jaffé, P.R., 2015. Characterization of incubation experiments and development of an enrichment culture capable of ammonium oxidation under iron-reducing conditions. Biogeosciences 12, 769–779. https://doi.org/10.5194/bg-12-769-2015.

Huang, S., Zhang, J., Wang, C., Zhu, G., Hassan, M., 2022. Weak electric field effect of MFC biocathode on denitrification. J. Environ. Chem. Eng. 10, 108596. https://doi.org/10.1016/j.jece.2022.108596.

Imachi, H., Sakai, S., Ohashi, A., Harada, H., Hanada, S., Kamagata, Y., Sekiguchi, Y., 2007. *Pelotomaculum propionicicum* sp. nov., an anaerobic, mesophilic, obligately syntrophic, propionate-oxidizing bacterium. Int. J. Syst. Evol. Microbiol. 57, 1487–1492. https://doi.org/10.1099/ijs.0.64925-0.

Janicek, A., Fan, Y., Liu, H., 2014. Design of microbial fuel cells for practical application: a review and analysis of scale-up studies. Biofuels 5, 79–92. https://doi.org/10.4155/bfs.13.69.

Jiang, M., Wang, P., Liu, H., Dai, X., Song, S., Liu, Y., 2022. The effect of operating strategies on the anaerobic digestion of gentamicin mycelial residues: insights into the enhancement of methane production and attenuation of gentamicin resistance. Environ. Sci. Technol. 56, 15130–15140. https://doi.org/10.1021/acs.est.2c00481.

Kato, S., Sasaki, K., Watanabe, K., Yumoto, I., Kamagata, Y., 2014. Physiological and transcriptomic analyses of the thermophilic, aceticlastic methanogen *Methanosaeta thermophila* responding to ammonia stress. Microb. Environ. 29, 162–167. https://doi.org/10.1264/jsme2.ME14021.

Khadir, A., Haroun, B., Jang, E., Santoro, D., Walton, J., Al-Omari, A., Muller, C., Bell, K.Y., Parker, W., Nakhla, G., 2025a. Methane production and microbial adaptation in high-load vacuum-enhanced anaerobic digestion: addressing ammonia and propionate toxicity. Chem. Eng. J. 509, 161105. https://doi.org/10.1016/j.cej.2025.161105.

Khadir, A., Jang, E., Santoro, D., Walton, J., Al-Omari, A., Muller, C., Bell, K.Y., Parker, W., Wells, M.J.M., Nakhla, G., 2026. High-rate vacuum-enhanced anaerobic digestion: performance, ammonia recovery and process kinetics. Chem. Eng. Sci. 320, 122475. https://doi.org/10.1016/j.ces.2025.122475.

Khadir, A., Jang, E., Santoro, D., Walton, J., Al-Omari, A., Muller, C., Bell, K.Y., Wells, M.J.M., Parker, W., Nakhla, G., 2024. Development of an innovative technology using side stream vacuum evaporation for anaerobic digestion intensification and ammonia recovery. Chem. Eng. J. 484, 149469. https://doi.org/10.1016/j.cej.2024.149469.

Khadir, A., Nakhla, G., Karki, R., Raskin, L., Muller, C., Guevarra, K., Summers, A., Pierce, L., Shahbaz, P., Bell, K., Skerlos, S., Bronstad, E., 2025b. Micro-aeration for hydrogen sulfide reduction in full-scale anaerobic digesters with limited headspace: performance and sulfide reduction kinetics. Process Saf. Environ. Prot. 196, 106911. https://doi.org/10.1016/j.psep.2025.106911.

Khanh Nguyen, V., Kumar Chaudhary, D., Hari Dahal, R., Hoang Trinh, N., Kim, J., Chang, S.W., Hong, Y., Duc La, D., Nguyen, X.C., Hao Ngo, H., Chung, W.J., Nguyen, D.D., 2021. Review on pretreatment techniques to improve anaerobic digestion of sewage sludge. Fuel 285, 119105. https://doi.org/10.1016/j.fuel.2020.119105.

Kim, G.-B., Cayetano, R.D.A., Park, J., Jo, Y., Park, S., Ersahin, M.E., Kim, S.-H., 2024. Effects of solids concentration and thermal pretreatment on continuous digestion of undigested dewatered sludge. Renew. Energy 231, 120894. https://doi.org/10.1016/j.renene.2024.120894.

Kosaka, T., Uchiyama, T., Ishii, S., Enoki, M., Imachi, H., Kamagata, Y., Ohashi, A., Harada, H., Ikenaga, H., Watanabe, K., 2006. Reconstruction and regulation of the central catabolic pathway in the thermophilic propionate-oxidizing syntroph *Pelotomaculum thermopropionicum*. J. Bacteriol. 188, 202–210. https://doi.org/10.1128/JB.188.1.202-210.2006.

Kunatsa, T., Xia, X., 2022. A review on anaerobic digestion with focus on the role of biomass co-digestion, modelling and optimisation on biogas production and enhancement. Bioresour. Technol. 344, 126311. https://doi.org/10.1016/j.biortech.2021.126311.

Kundu, R., Kunnoth, B., Pilli, S., Polisetty, V.R., Tyagi, R.D., 2023. Biochar symbiosis in anaerobic digestion to enhance biogas production: a comprehensive review. J. Environ. Manag. 344, 118743. https://doi.org/10.1016/j.jenvman.2023.118743.

Lebuhn, M., Weiß, S., Munk, B., Guebitz, G.M., 2015. Microbiology and molecular biology tools for biogas process analysis. Diagnosis and Control 1–40. https://doi.org/10.1007/978-3-319-21993-6_1.

Lee, M.J., Zinder, S.H., 1988. Isolation and characterization of a thermophilic bacterium which oxidizes acetate in syntrophic association with a methanogen and which grows acetogenically on H_2-CO_2. Appl. Environ. Microbiol. 54, 124–129. https://doi.org/10.1128/aem.54.1.124-129.1988.

Li, C., Lü, F., Peng, W., He, P., Zhang, H., 2024. Efficacy of bioaugmentation with nondomesticated mixed microbial consortia under ammonia inhibition in anaerobic digestion. Bioresour. Technol. 391, 129954. https://doi.org/10.1016/j.biortech.2023.129954.

Li, Z.-Y., Inoue, D., Ike, M., 2023. Mitigating ammonia-inhibition in anaerobic digestion by bioaugmentation: a review. J. Water Process Eng. 52, 103506. https://doi.org/10.1016/j.jwpe.2023.103506.

Liu, C., Huang, H., Duan, X., Chen, Y., 2021. Integrated metagenomic and metaproteomic analyses unravel ammonia toxicity to active methanogens and syntrophs, enzyme synthesis, and key enzymes in anaerobic digestion. Environ. Sci. Technol. 55, 14817–14827. https://doi.org/10.1021/acs.est.1c00797.

Liu, C., Zhang, T., Li, L., Liu, S., Bo, J., Mouhib, M., He, Q., 2024. Non-targeted facilitation of primary sludge anaerobic fermentation by micro-aeration and the simultaneous nutrients transformations. Chem. Eng. J. 491, 151930. https://doi.org/10.1016/j.cej.2024.151930.

Liu, S., Luo, S., Li, L., Han, L., Ye, W., Peng, X., 2025. Coupling applied voltage with high-solids anaerobic digestion to achieve synchronous nitrogen removal and methanogenesis: performance and mechanisms. J. Environ. Chem. Eng. 13, 119925. https://doi.org/10.1016/j.jece.2025.119925.

Liu, Y., Balkwill, D.L., Aldrich, H.C., Drake, G.R., Boone, D.R., 1999. Characterization of the anaerobic propionate-degrading syntrophs *Smithella propionica* gen. nov., sp. nov. and *Syntrophobacter wolinii*. Int. J. Syst. Evol. Microbiol. 49, 545–556. https://doi.org/10.1099/00207713-49-2-545.

Liu, Y., Ying, L., Li, H., Awasthi, M.K., Tian, D., He, J., Zou, J., Lei, Y., Shen, F., 2024. Allophane improves anaerobic digestion of chicken manure by alleviating ammonia inhibition and intensifying direct interspecies electron transfer. Bioresour. Technol. 400, 130692. https://doi.org/10.1016/j.biortech.2024.130692.

Ma, J., Pan, J., Zhang, Y., Yao, Z., Yu, J., Luo, J., Shen, R., Awasthi, M.K., Zhao, L., 2022. Alleviating "inhibited steady-state" in anaerobic digestion of poultry manure by bentonite amendment: performance evaluation and microbial mechanism. Bioresour. Technol. 360, 127519. https://doi.org/10.1016/j.biortech.2022.127519.

Manzo, R., Papirio, S., Esposito, G., Matassa, S., 2026. Enabling protein-rich microalgae production from anaerobic digestate through direct aeration stripping for gaseous nitrogen and carbon supply. Waste Manag. 209, 115187. https://doi.org/10.1016/j.wasman.2025.115187.

Moerland, M.J., Bruning, H., Buisman, C.J.N., van Eekert, M.H.A., 2021. Advanced modelling to determine free ammonia concentrations during (hyper-)thermophilic anaerobic digestion in high strength wastewaters. J. Environ. Chem. Eng. 9, 106724. https://doi.org/10.1016/j.jece.2021.106724.

Mustafa Abdelrahman, A., Khadir, A., Santoro, D., Jang, E., Al-Omari, A., Muller, C., Bell, K.Y., Walton, J., Batstone, D., Nakhla, G., 2025. Vacuum evaporation coupled with anaerobic digestion for process intensification and ammonia recovery: model development, validation and scenario analysis. Bioresour. Technol. 416, 131753. https://doi.org/10.1016/j.biortech.2024.131753.

Nielsen, H.B., Angelidaki, I., 2008. Co-digestion of manure and industrial organic waste at centralized biogas plants: process imbalances and limitations. Water Sci. Technol. 58, 1521–1528. https://doi.org/10.2166/wst.2008.507.

Nilsen, R.K., Torsvik, T., Lien, T., 1996. *Desulfotomaculum thermocisternum* sp. nov., a sulfate reducer isolated from a hot North Sea oil reservoir. Int. J. Syst. Bacteriol. 46, 397–402. https://doi.org/10.1099/00207713-46-2-397.

Palakodeti, A., Azman, S., Rossi, B., Dewil, R., Appels, L., 2021. A critical review of ammonia recovery from anaerobic digestate of organic wastes via stripping. Renew. Sustain. Energy Rev. 143, 110903. https://doi.org/10.1016/j.rser.2021.110903.

Pant, D., Singh, A., Van Bogaert, G., Gallego, Y.A., Diels, L., Vanbroekhoven, K., 2011. An introduction to the life cycle assessment (LCA) of bioelectrochemical systems (BES) for sustainable energy and product generation: relevance and key aspects. Renew. Sustain. Energy Rev. 15, 1305–1313. https://doi.org/10.1016/j.rser.2010.10.005.

Peng, Y., Li, L., Yang, P., Liu, H., Ye, W., Xue, Z., Peng, X., Wang, X., 2023. Integrated genome-centric metagenomic and metaproteomic analyses unravel the responses of the microbial community to ammonia stress. Water Res. 242, 120239. https://doi.org/10.1016/j.watres.2023.120239.

Peng, Y., Li, L., Yuan, W., Wu, D., Yang, P., Peng, X., 2022. Long-term evaluation of the anaerobic co-digestion of food waste and landfill leachate to alleviate ammonia inhibition. Energy Convers. Manag. 270, 116195. https://doi.org/10.1016/j.enconman.2022.116195.

Peng, Y., Yang, P., Zhang, Y., Wang, X., Peng, X., Li, L., 2021. Consecutive batch anaerobic digestion under ammonia stress: microbial community assembly and process performance. J. Environ. Chem. Eng. 9, 106061. https://doi.org/10.1016/j.jece.2021.106061.

Plugge, C.M., Balk, M., Stams, A.J.M., 2002. Desulfotomaculum thermobenzoicum subsp. thermosyntrophicum subsp. nov., a thermophilic, syntrophic, propionate-oxidizing, spore-forming bacterium. Int. J. Syst. Evol. Microbiol. 52, 391–399. https://doi.org/10.1099/00207713-52-2-391.

Plugge, C.M., van Lier, J.B., Stams, A.J.M., 2010. Syntrophic communities in methane formation from high strength wastewaters. In: Microbes at Work. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 59–77. https://doi.org/10.1007/978-3-642-04043-6_3.

Primmer, N., 2021. Biogas: Pathways to 2030.

Romero-Güiza, M.S., Vila, J., Mata-Alvarez, J., Chimenos, J.M., Astals, S., 2016. The role of additives on anaerobic digestion: a review. Renew. Sustain. Energy Rev. 58, 1486–1499. https://doi.org/10.1016/j.rser.2015.12.094.

Ruan, D., Zhou, Z., Pang, H., Yao, J., Chen, G., Qiu, Z., 2019. Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. Bioresour. Technol. 289, 121643. https://doi.org/10.1016/j.biortech.2019.121643.

Rubin, B.E., Diamond, S., Cress, B.F., Crits-Christoph, A., Lou, Y.C., Borges, A.L., Shivram, H., He, C., Xu, M., Zhou, Z., Smith, S.J., Rovinsky, R., Smock, D.C.J., Tang, K., Owens, T.K., Krishnappa, N., Sachdeva, R., Barrangou, R., Deutschbauer, A. M., Banfield, J.F., Doudna, J.A., 2021. Species- and site-specific genome editing in complex bacterial communities. Nat. Microbiol. 7, 34–47. https://doi.org/10.1038/s41564-021-01014-7.

Schnurer, A., Schink, B., Svensson, B.H., 1996. *Clostridium ultunense* sp. nov., a mesophilic bacterium oxidizing acetate in syntrophic association with a hydrogenotrophic methanogenic bacterium. Int. J. Syst. Bacteriol. 46, 1145–1152. https://doi.org/10.1099/00207713-46-4-1145.

Shi, K., Ju, J., Elsamadony, M., Fujii, M., Liu, J., Qin, J., Liao, Z., Ou, C., 2024. Emerging feammox technology: mechanisms, biotechnological applications, and future prospects. ACS ES&T Eng. 4, 2856–2873. https://doi.org/10.1021/acsestengg.4c00525.

Shrestha, J., Rich, J.J., Ehrenfeld, J.G., Jaffe, P.R., 2009. Oxidation of ammonium to nitrite under iron-reducing conditions in wetland soils. Soil Sci. 174, 156–164. https://doi.org/10.1097/SS.0b013e3181988fbf.

Singh, A., Schnürer, A., Dolfing, J., Westerholm, M., 2023. Syntrophic entanglements for propionate and acetate oxidation under thermophilic and high-ammonia conditions. ISME J. 17, 1966–1978. https://doi.org/10.1038/s41396-023-01504-y.

Singh, A., Schnürer, A., Westerholm, M., 2021. Enrichment and description of novel bacteria performing syntrophic propionate oxidation at high ammonia level. Environ. Microbiol. 23, 1620–1637. https://doi.org/10.1111/1462-2920.15388.

Sorokin, D.Y., Chernyh, N.A., 2016. '*Candidatus Desulfonatronobulbus propionicus*': a first haloalkaliphilic member of the order Syntrophobacterales from soda lakes. Extremophiles 20, 895–901. https://doi.org/10.1007/s00792-016-0881-3.

Sun, L., Müller, B., Westerholm, M., Schnürer, A., 2014. Syntrophic acetate oxidation in industrial CSTR biogas digesters. J. Biotechnol. 171, 39–44. https://doi.org/10.1016/j.jbiotec.2013.11.016.

Tang, K.H.D., Lock, S.S.M., Yap, P.-S., Cheah, K.W., Chan, Y.H., Yiin, C.L., Ku, A.Z.E., Loy, A.C.M., Chin, B.L.F., Chai, Y.H., 2022. Immobilized enzyme/microorganism complexes for degradation of microplastics: a review of recent advances, feasibility and future prospects. Sci. Total Environ. 832, 154868. https://doi.org/10.1016/j.scitotenv.2022.154868.

Thiel, P.G., Hattingh, W.H.J., 1967. Determination of hydrolytic enzyme activities in anaerobic digesting sludge. Water Res. 1, 191–196. https://doi.org/10.1016/0043-1354(67)90009-7.

Wallrabenstein, C., Hauschild, E., Schink, B., 1995. *Syntrophobacter pfennigii* sp. nov., new syntrophically propionate-oxidizing anaerobe growing in pure culture with propionate and sulfate. Arch. Microbiol. 164, 346–352. https://doi.org/10.1007/s002030050273.

Wang, C., Liu, J., Xu, X., Zhu, L., 2022. Response of methanogenic granules enhanced by magnetite to ammonia stress. Water Res. 212, 118123. https://doi.org/10.1016/j.watres.2022.118123.

Wang, D., Duan, Y., Yang, Q., Liu, Y., Ni, B.-J., Wang, Q., Zeng, G., Li, X., Yuan, Z., 2018. Free ammonia enhances dark fermentative hydrogen production from waste activated sludge. Water Res. 133, 272–281. https://doi.org/10.1016/j.watres.2018.01.051.

Wang, G., Fu, P., Zhang, B., Zhang, Ji, Huang, Q., Yao, G., Li, Q., Dzakpasu, M., Zhang, Jianfeng, Li, Y.-Y., Chen, R., 2023. Biochar facilitates methanogens evolution by enhancing extracellular electron transfer to boost anaerobic digestion of swine manure under ammonia stress. Bioresour. Technol. 388, 129773. https://doi.org/10.1016/j.biortech.2023.129773.

Wang, H., Fotidis, I.A., Angelidaki, I., 2015. Ammonia effect on hydrogenotrophic methanogens and syntrophic acetate-oxidizing bacteria. FEMS Microbiol. Ecol. 91, 130. https://doi.org/10.1093/femsec/fiv130.

Wang, H., Zheng, X., Yan, Q., Zhang, G., Kim, J.R., 2021. Microbial community and metabolic responses to electrical field intensity for alleviation of ammonia inhibition in an integrated bioelectrochemical system (BES). Bioresour. Technol. 336, 125332. https://doi.org/10.1016/j.biortech.2021.125332.

Wang, S., Wang, Z., Usman, M., Zheng, Z., Zhao, X., Meng, X., Hu, K., Shen, X., Wang, X., Cai, Y., 2023. Two microbial consortia obtained through purposive acclimatization as biological additives to relieve ammonia inhibition in anaerobic digestion. Water Res. 230, 119583. https://doi.org/10.1016/j.watres.2023.119583.

Wang, Y., Li, H., Ding, K., Zhao, X., Liu, M., Xu, L., Gu, L., Li, J., Li, L., He, Q., Liang, J., 2025. Improved anaerobic digestion of food waste under ammonia stress by side-stream hydrogen domestication. Water Res. 268, 122770. https://doi.org/10.1016/j.watres.2024.122770.

Wang, Z., He, H., Yan, J., Xu, Z., Yang, G., Wang, H., Zhao, Y., Cui, Z., Yuan, X., 2024. Influence of temperature fluctuations on anaerobic digestion: optimum performance is achieved at 45 °C. Chem. Eng. J. 492, 152331. https://doi.org/10.1016/j.cej.2024.152331.

Wei, C.-H., Wang, Z.-W., Dai, J.-H., Xiao, K., Yu, H.-R., Qu, F.-S., Rong, H.-W., He, J.-G., Ngo, H.H., 2024. Enhanced anaerobic digestion performance and sludge filterability by membrane microaeration for anaerobic membrane bioreactor application. Bioresour. Technol. 402, 130787. https://doi.org/10.1016/j.biortech.2024.130787.

Westerholm, M., Dolfing, J., Sherry, A., Gray, N.D., Head, I.M., Schnürer, A., 2011a. Quantification of syntrophic acetate-oxidizing microbial communities in biogas processes. Environ. Microbiol. Rep. 3, 500–505. https://doi.org/10.1111/j.1758-2229.2011.00249.x.

Westerholm, M., Müller, B., Isaksson, S., Schnürer, A., 2015. Trace element and temperature effects on microbial communities and links to biogas digester performance at high ammonia levels. Biotechnol. Biofuels 8, 154. https://doi.org/10.1186/s13068-015-0328-6.

Westerholm, M., Roos, S., Schnürer, A., 2011b. *Tepidanaerobacter acetatoxydans* sp. nov., an anaerobic, syntrophic acetate-oxidizing bacterium isolated from two ammonium-enriched mesophilic methanogenic processes. Syst. Appl. Microbiol. 34, 260–266. https://doi.org/10.1016/j.syapm.2010.11.018.

Westerholm, M., Roos, S., Schnürer, A., 2010. *Syntrophaceticus schinkii* gen. nov., sp. nov., an anaerobic, syntrophic acetate-oxidizing bacterium isolated from a mesophilic anaerobic filter. FEMS Microbiol. Lett. https://doi.org/10.1111/j.1574-6968.2010.02023.x no-no.

Wett, B., Takács, I., Batstone, D., Wilson, C., Murthy, S., 2014. Anaerobic model for high-solids or high-temperature digestion – additional pathway of acetate oxidation. Water Sci. Technol. 69, 1634–1640. https://doi.org/10.2166/wst.2014.047.

Wu, B., Lin, R., Gu, J., Yuan, H., Murphy, J.D., 2024. Biochar confers significant microbial resistance to ammonia toxicity in n-caproic acid production. Water Res. 266, 122367. https://doi.org/10.1016/j.watres.2024.122367.

Wu, L.-J., Li, X.-X., Ye, F., Liu, Y.-X., Yang, F., Zhou, Q., Lyu, Y.-K., 2024. Ammonia stripping by in situ biogas self-circulation to upgrade continuous thermophilic and mesophilic digestion of hydrothermal high-solid sludge. Bioresour. Technol. 402, 130797. https://doi.org/10.1016/j.biortech.2024.130797.

Wu, Y., Du, Q., Wan, Y., Zhao, Q., Li, N., Wang, X., 2022. Autotrophic nitrate reduction to ammonium via reverse electron transfer in *Geobacter* dominated biofilm. Biosens. Bioelectron. 215, 114578. https://doi.org/10.1016/j.bios.2022.114578.

Xia, Q., Ai, Z., Huang, Wenli, Yang, F., Liu, F., Lei, Z., Huang, Weiwei, 2022. Recent progress in applications of Feammox technology for nitrogen removal from wastewaters: a review. Bioresour. Technol. 362, 127868. https://doi.org/10.1016/j.biortech.2022.127868.

Xia, Q., Liu, F., Sun, S., Huang, Wenli, Zhao, Z., Yang, F., Lei, Z., Huang, Weiwei, Yi, X., 2023. Coupling iron sludge addition and intermittent aeration for achieving simultaneous methanogenesis, feammox, and denitrification in a single reactor treating fish sludge. Environ. Sci. Technol. 57, 15065–15075. https://doi.org/10.1021/acs.est.3c03009.

Xing, J., Song, J., Liu, C., Yang, W., Duan, H., Yabar, H., Ren, J., 2022. Integrated crop–livestock–bioenergy system brings co-benefits and trade-offs in mitigating the environmental impacts of Chinese agriculture. Nat. Food 3, 1052–1064. https://doi.org/10.1038/s43016-022-00649-x.

Xu, J., Xu, L., Zong, Y., Lin, R., He, Y., Xie, L., 2024. Electrically-assisted anaerobic digestion under ammonia stress: facilitating propionate oxidation and activating methanogenesis via direct interspecies electron transfer. Bioresour. Technol. 393, 130067. https://doi.org/10.1016/j.biortech.2023.130067.

Yamamoto-Ikemoto, R., Matsuura, N., Honda, R., Hara-Yamamura, H., Some, K., Prak, S., Koike, K., Togari, T., 2023. Ammonia tolerance and microbial community in thermophilic co-digestion of sewage sludge initiated with lignocellulosic biomass. Bioresour. Technol. 376, 128834. https://doi.org/10.1016/j.biortech.2023.128834.

Yang, S., Chen, Z., Wen, Q., 2021. Impacts of biochar on anaerobic digestion of swine manure: methanogenesis and antibiotic resistance genes dissemination. Bioresour. Technol. 324, 124679. https://doi.org/10.1016/j.biortech.2021.124679.

Yang, Y., Xiao, C., Lu, J., Zhang, Y., 2020. Fe(III)/Fe(II) forwarding a new anammox-like process to remove high-concentration ammonium using nitrate as terminal electron acceptor. Water Res. 172, 115528. https://doi.org/10.1016/j.watres.2020.115528.

Yang, Y., Xiao, C., Yu, Q., Zhao, Z., Zhang, Y., 2021a. Using Fe(II)/Fe(III) as catalyst to drive a novel anammox process with no need of anammox bacteria. Water Res. 189, 116626. https://doi.org/10.1016/j.watres.2020.116626.

Yang, Y., Zhang, Y., Li, Y., Zhao, H., Peng, H., 2018. Nitrogen removal during anaerobic digestion of wasted activated sludge under supplementing Fe(III) compounds. Chem. Eng. J. 332, 711–716. https://doi.org/10.1016/j.cej.2017.09.133.

Yang, Y., Zhao, Z., Zhang, Y., 2021b. Anaerobic ammonium removal pathway driven by the Fe(II)/Fe(III) cycle through intermittent aeration. Environ. Sci. Technol. 55, 7615–7623. https://doi.org/10.1021/acs.est.0c08624.

Yang, Z., Sun, H., Zhou, L., Arhin, S.G., Papadakis, V.G., Goula, M.A., Liu, G., Zhang, Y., Wang, W., 2022. Bioaugmentation with well-constructed consortia can effectively alleviate ammonia inhibition of practical manure anaerobic digestion. Water Res. 215, 118244. https://doi.org/10.1016/j.watres.2022.118244.

Yellezuome, D., Zhu, X., Wang, Z., Liu, R., 2022. Mitigation of ammonia inhibition in anaerobic digestion of nitrogen-rich substrates for biogas production by ammonia stripping: a review. Renew. Sustain. Energy Rev. 157, 112043. https://doi.org/10.1016/j.rser.2021.112043.

Yenigün, O., Demirel, B., 2013. Ammonia inhibition in anaerobic digestion: a review. Process Biochemistry 48, 901–911. https://doi.org/10.1016/j.procbio.2013.04.012.

Yi, Y., Wang, H., Chen, Y., Gou, M., Xia, Z., Hu, B., Nie, Y., Tang, Y., 2020. Identification of novel Butyrate- and acetate-oxidizing bacteria in butyrate-fed mesophilic anaerobic chemostats by DNA-based stable isotope probing. Microb. Ecol. 79, 285–298. https://doi.org/10.1007/s00248-019-01400-z.

Yin, Q., Gu, M., Wu, G., 2020. Inhibition mitigation of methanogenesis processes by conductive materials: a critical review. Bioresour. Technol. 317, 123977. https://doi.org/10.1016/j.biortech.2020.123977.

Yu, J., Usman, M., Liu, F., Schäfer, F., Shen, Y., Zheng, Z., Cai, Y., 2025. CO2 agitation combined with magnetized biochar to alleviate "ammonia inhibited steady-state": exploring the mechanism by combining metagenomics with macroscopic indicators. Water Res. 276, 123250. https://doi.org/10.1016/j.watres.2025.123250.

Yu, X., Wei, F., Wang, H., Bi, S., Zhao, C., Wang, Y., 2025. Enhancement of high-solid chicken manure anaerobic digestion: insights into ammonia air stripping pretreatment for microbial resilience and hydraulic retention stabilization. Biomass Bioenergy 201, 108098. https://doi.org/10.1016/j.biombioe.2025.108098.

Yuan, T., Sun, R., Miao, Q., Wang, X., Xu, Q., 2023. Analysing the mechanism of food waste anaerobic digestion enhanced by iron oxide in a continuous two-stage process. Waste Manag. 171, 610–620. https://doi.org/10.1016/j.wasman.2023.10.009.

Zhang, H., Peng, Y., Yang, P., Wang, X., Peng, X., Li, L., 2020. Response of process performance and microbial community to ammonia stress in series batch experiments. Bioresour. Technol. 314, 123768. https://doi.org/10.1016/j.biortech.2020.123768.

Zhang, H., Yuan, W., Dong, Q., Wu, D., Yang, P., Peng, Y., Li, L., Peng, X., 2022. Integrated multi-omics analyses reveal the key microbial phylotypes affecting anaerobic digestion performance under ammonia stress. Water Res. 213, 118152. https://doi.org/10.1016/j.watres.2022.118152.

Zhang, J., Zhu, G., Lv, N., Pan, X., Li, L., Ren, Z.J., 2017. The establishment and characteristics of dominant syntrophic propionate oxidation bacteria and sulfate-reducing bacteria in a mixed culture. Chem. Eng. Commun. 204, 926–936. https://doi.org/10.1080/00986445.2017.1328410.

Zhang, S., Ma, X., Xie, D., Guan, W., Yang, M., Zhao, P., Gao, M., Wang, Q., Wu, C., 2021. Adding activated carbon to the system with added zero-valent iron further improves anaerobic digestion performance by alleviating ammonia inhibition and promoting DIET. J. Environ. Chem. Eng. 9, 106616. https://doi.org/10.1016/j.jece.2021.106616.

Zhang, S., Xiao, M., Liang, C., Chui, C., Wang, N., Shi, J., Liu, L., 2022. Multivariate insights into enhanced biogas production in thermophilic dry anaerobic co-digestion of food waste with kitchen waste or garden waste: process properties, microbial communities and metagenomic analyses. Bioresour. Technol. 361, 127684. https://doi.org/10.1016/j.biortech.2022.127684.

Zheng, Y., Wang, J., Niu, X., Su, X., He, X., Hu, Y., Zhao, D., Chen, D., Lin, Y., Li, K., 2025a. Enhancement of methane production from pig manure by ferrate pretreated anaerobic digestion: performance and mechanism. Chem. Eng. J. 515, 163862. https://doi.org/10.1016/j.cej.2025.163862.

Zheng, Y., Wang, J., Niu, X., Su, X., He, X., Hu, Y., Zhao, D., Chen, D., Lin, Y., Li, K., 2025b. Ferrihydrite enhance performance in anaerobic digestion of pig manure: methane production, Feammox and metabolic pathway. J. Water Process Eng. 72, 107621. https://doi.org/10.1016/j.jwpe.2025.107621.

Zheng, Y., Yan, C., Niu, X., Liu, M., Su, X., He, X., Hu, Y., Zhao, D., Li, K., 2025c. Iron (hydr) oxides enhance anaerobic digestion of swine manure: methane production and iron ammonia oxidation. J. Environ. Chem. Eng. 13, 118582. https://doi.org/10.1016/j.jece.2025.118582.

Zheng, Z., Cai, Y., Zhang, Y., Zhao, Y., Gao, Y., Cui, Z., Hu, Y., Wang, X., 2021. The effects of C/N (10–25) on the relationship of substrates, metabolites, and microorganisms in "inhibited steady-state" of anaerobic digestion. Water Res. 188, 116466. https://doi.org/10.1016/j.watres.2020.116466.

Zhou, J., Ming, S., Liu, Q., Zhang, Y., Duan, N., 2024. Revealing the synergy mechanisms of organic components anaerobic co-digestion from the prevailing tendency of endogenous inhibitors. Chem. Eng. J. 479, 147707. https://doi.org/10.1016/j.cej.2023.147707.

Zinder, S.H., Koch, M., 1984. Non-aceticlastic methanogenesis from acetate: acetate oxidation by a thermophilic syntrophic coculture. Arch. Microbiol. 138, 263–272. https://doi.org/10.1007/BF00402133.